

DATA MANIPULATION WITH PYTHON

Fundamentals, Numpy, Pandas

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2026-III



Outline

- 1 Foundations of Python
- 2 Numerical Analysis with Numpy
- 3 Data Manipulation with Pandas



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Basic Definitions in Python

while for if

- **Python** is a high-level, interpreted, and general-purpose programming language.
- **High-level** means that Python is designed to be easy to read and write.
- **Interpreted** means that Python code is executed line by line, rather than being compiled into machine code.
- **Weakly typed** means that Python does not require you to declare the type of a variable.

Processing → UI / Graphic

- **Multiparadigm** means that Python supports object-oriented, imperative, and functional programming styles.
- **Snake-case** is the convention of writing variable names in lowercase, with underscores between words.



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1st

pycode/main.py

main.py



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Java → myVar (camel)

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my_var

my_function



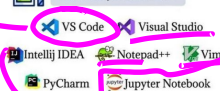
Popular Developer Tools

Life is Short, Use Dev Tools

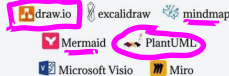
ByteByteGo

FOSS

Development Env



Diagramming



AI Tools



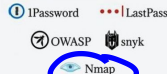
Hosting & Deployment



Code Quality



Security



Note-taking



Design



Virtual Environments

- **Virtual environments** are a way to create isolated spaces on your computer for **Python projects**.
- **Virtual environments** allow you to **install packages** and **dependencies** for a specific project without affecting other projects.

Handwritten notes illustrating virtual environments:

2017 | Python 2
| Python 3
2022 ;

Box 1 (Red border):
Pr. 1
Python 2.7
Numpy 1.2

Box 2 (Red border):
Pr. 2
Python 3.11
Numpy 2.0

virtual on

virtual on



Virtual Environments

- **Virtual environments** are a way to create isolated spaces on your computer for **Python projects**.
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- **Virtual environments** are created using the **venv** module, which is included in the **Python** standard library.
- **Virtual environments** are activated using the **source** command in the terminal.
- **Virtual environments** are deactivated using the **deactivate** command in the terminal.



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(venv) ~ - \$? } terminal



Modules and Packages

- **Modules** are Python files that contain functions, classes, and variables.
- **Modules** are used to organize code and make it reusable.
- Packages are directories that contain Python files (modules).
- Packages are used to organize modules and make them reusable.

→ all files with
extension .py



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✱ name_folder/
--init--.py



Dependencies Management

- **Dependencies** are external libraries or packages that your Python project relies on.
- **Dependencies** are traditionally managed using a pyproject.toml file, which lists the names and versions of the packages required by your project.
- **Dependencies** can be installed using the pip package manager, which is included with Python.
- Currently, the poetry package manager is recommended for managing dependencies in Python projects. It follows the PEP 508 standard for specifying dependencies.

old:

requirements.txt

standard



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Jupyter Notebooks

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- **Jupyter Notebooks** support *multiple programming languages*, including **Python**, **R**, and **Julia**.

- W. Illyg Carol
fernandez

COLAB (Google)

Jupyter



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- **Jupyter Notebooks** are used for *data cleaning*, *data transformation*, *statistical modeling*, *data visualization*, *machine learning*, and *more*.
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Jupyter Notebooks

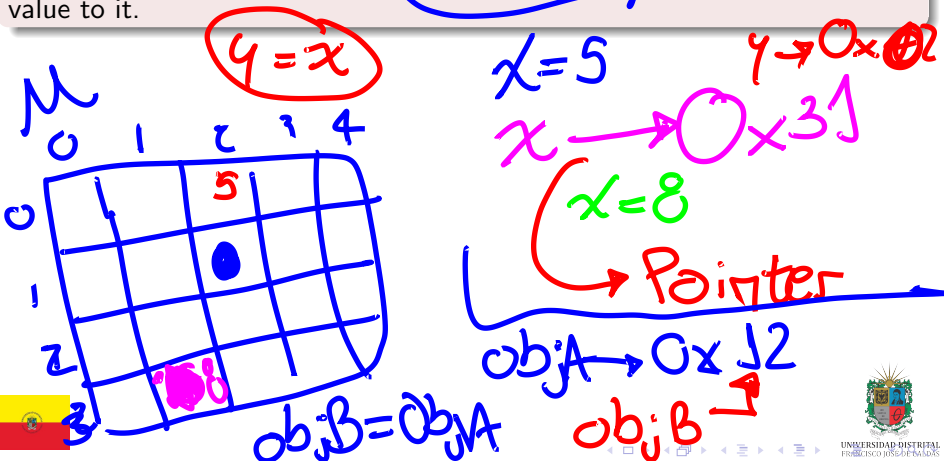
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Variables and Memory Management

Variables Definition

Variables are used to store data values. Python has no command for declaring a variable. A variable is created the moment you first assign a value to it.



Conditionals

Definition

Conditionals are used to execute different code blocks based on *different conditions*.

Nested Conditionals

Nested conditionals are conditionals that are *inside other conditionals*.

Elif Conditionals

Elif conditionals are used to check *multiple conditions*.

question → *analogic logic*

Yes No



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Question:

Yes →

Question

No → Question



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Question
 " " " " " "
 Case 1?
 Case 2?
 Case 3?

alternatives
for the
same variable



Loops and Range

Loops Definition

Loops are used to execute a block of code *multiple times*.

Range Definition

The range function is used to generate a sequence of numbers.

while

for

- executions: ?

- executions: 17



Loops and Range

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Loops are used to execute a block of code *multiple times*.

Range Definition

The **range function** is used to generate a **sequence** of numbers.

sequence - series

and exc.
range(10): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

start, end exc.
range(5, 9) = 5, 6, 7, 8

range(10, 0, -1) = 9, 8, 7, 6, 5, 4, 3, 2, 1



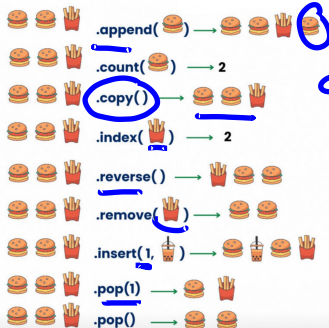
Lists → no arrays

Definition

A **list** is a collection of **items** that are *ordered* and *changeable*. **Lists** are defined by enclosing the items in square brackets.

obj A = obj B
memory

PYTHON LIST METHODS



list = []

obj A copy obj B
m1 m2



Dictionaries

Definition

A **dictionary** is a collection of items that are *unordered*, *changeable*, and *indexed*. Dictionaries are defined by enclosing the items in curly braces.



Sets and Tuples

Definition Sets

A **set** is a collection of items that are unordered and unindexed. **Sets** are defined by enclosing the items in curly braces.

Definition Tuples

A **tuple** is a collection of items that are ordered and unchangeable. **Tuples** are defined by enclosing the items in parentheses.

Handwritten notes:

- set = {1, 2, 3}* (with red text: "no duplicates", "Union", "Intersection")
- dict = {'a': 1, 'b': 2}*



Sets and Tuples

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Definition Tuples

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tuple = (1, 2, 3) ^{0 1 2 index}



Lists Comprehensions

Definition

List comprehensions provide a concise way to create lists. Common applications are to make *new lists* where each element is the result of some operation applied to each member of another sequence or iterable, or to create a subsequence of those elements that satisfy a certain condition.

$$\text{even} = [x \text{ for } x \text{ in range}(100) \text{ if } x \% 2 == 0]$$

$$\text{square} = [i^2 \text{ for } i \text{ in range}(1, 11)]$$

Handwritten annotations: In the first line, a red bracket under 'x in range(100)' is labeled with a circled '1', and a red bracket under 'if x % 2 == 0' is labeled with a circled '2'. In the second line, a pink bracket under 'i in range(1, 11)' is labeled with a circled '2', and a pink arrow points from this bracket to the 'i^2' term.



Functions

Define f as a function ~

Definition

A **function** is a block of code that only **runs** when it is called. You can pass data, known as **parameters**, into a function. A **function** can return data as a result.

Type of Functions

- Built-in Functions
- User-defined Functions
 - Variadic Functions
 - Recursive Functions

def $f(x)$:

$y = x + 2$

return y

parameters

type hints

def $f(x: \text{int}) \rightarrow \text{int}$:
 $y = x + 2$
 return y



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docstring → google style

""" explain

Angsi

defunsi



Python Iterators

Vectorization → no for

Definition

An **iterator** is an object that contains a countable number of values. An **iterator** is an object that can be iterated upon, meaning that you can traverse through all the values.

Maps

The **map function** is used to apply a function to *all the items* in an input list.

Filters

The **filter function** is used to *select items* from an input list that meet a certain condition.



Python Iterators

Definition

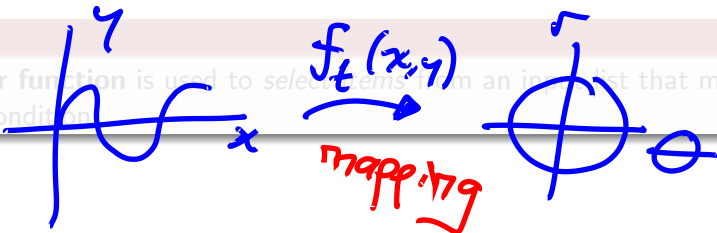
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el/3
e/2
e/3

Cond
T
F
T



el/3
e/3



Lambda Functions

Definition

A **lambda function** is a small **anonymous** function. A **lambda function** can take any number of arguments, but can only have one expression.

lambda x: x * x
 argument return



Classes and Objects

Definition

Python is an *object-oriented programming* language. Almost everything in Python is an **object**, with its *properties* and *methods*. A **class** is like an object constructor, or a “blueprint” for creating objects.

class A: all methods

def __init__(self): instance

pass

class B (A): inheritance

obj = A()



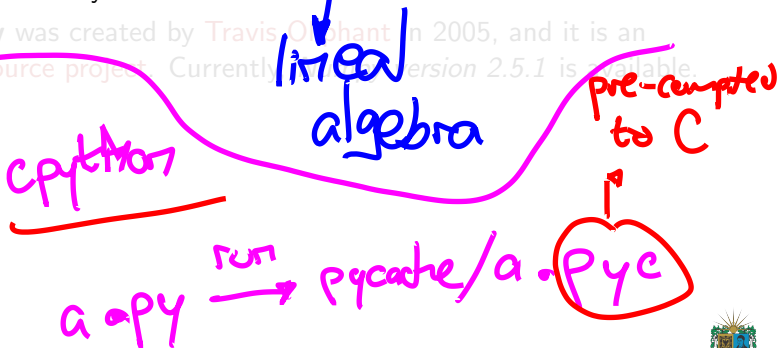
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Numerical Python Library — Numpy

- **Numpy** is the **core** library for **scientific computing** in **Python**. It is the **fundamental package** for scientific computing with **Python**.
- **Numpy** is a **general-purpose array-processing** package. It provides a **high-performance** multidimensional array object, and tools for working with these arrays.
- **Numpy** was created by **Travis Oliphant** in 2005, and it is an open-source project. Currently, **version 2.5.1** is available.



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numpy
scipy

Anaconda
Numfocus → Spyder IDE
Pydata
• TeamsCollab



Lineal Algebra with Numpy

- **Numpy** provides a *comprehensive* set of **linear algebra** functions.
 - **Numpy** provides the *functionality* to create and **manipulate matrices**.
 - **Numpy** provides the *functionality* to solve linear systems of equations.
 - **Numpy** provides the *functionality* to calculate the determinant of a matrix.
 - **Numpy** provides the *functionality* to calculate the inverse of a matrix.
- machine learning



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$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & \dots & \dots \\ a_{31} & \dots & \dots & \dots \\ \vdots & \dots & \dots & \dots \\ a_{m1} & \dots & \dots & a_{mn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\begin{array}{l|l} 2x + 1y = 1 & x = 0.43 \\ 3x + 5y = 2 & y = 0.14 \end{array}$$



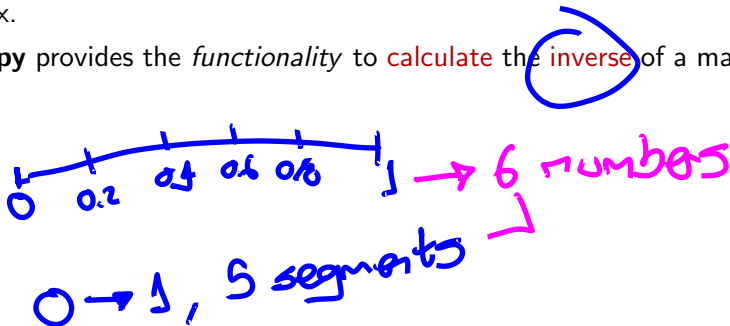
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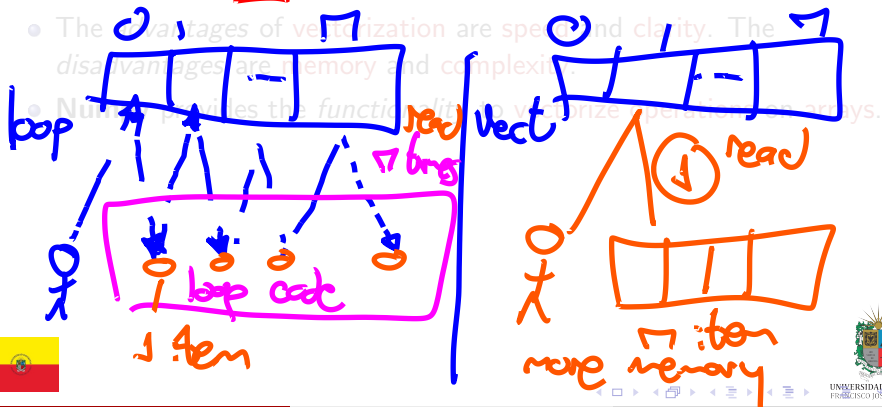
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Vectorization with Numpy → No loops

- **Vectorization** is the process of converting an algorithm from operating on a single value at a time to operating on a set of values at one time.
- **Vectorization** is the process of replacing explicit loops with array expressions or matrix operations.



Vectorization with Numpy

- **Vectorization** is the *process* of **converting** an **algorithm** from **operating** on a single value at a time to **operating** on a set of values at one time.
- **Vectorization** is the *process* of **replacing** explicit **loops** with **array expressions** or **matrix operations**. $0x - 300x$
- The *advantages* of **vectorization** are **speed** and **clarity**. The *disadvantages* are **memory** and **complexity**. $\rightarrow$ library
- **Numpy** provides the *functionality* to **vectorize operations** on **arrays**.



Typical Operations with Numpy

- **Numpy** provides the *functionality* to create and manipulate arrays.
- **Numpy** provides the *functionality* to perform element-wise operations on arrays, matrix operations on arrays.
- **Numpy** provides the *functionality* to perform linear algebra operations on arrays.
- **Numpy** provides the *functionality* to perform statistical operations on arrays.



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Introduction to Pandas

- **Pandas** is a *fast, powerful, flexible*, and *easy-to-use* *open-source* data *manipulation* and *data analysis* library built on top of the Python programming language.
- **Pandas** is a *high-level* data manipulation tool developed by *Wes McKinney* in *2008*. → *Matt Harrison*
- Pandas is a fast and efficient data manipulation tool that is built on top of NumPy.
- Pandas is one of the most popular and widely used data manipulation libraries in the world.

Pandas in an excel into Python



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→ CPython

→ Rust → Polars
≈ C



The "Series" Data Structure

- A **Series** is a **one-dimensional array-like object** that contains a **sequence of values** and an associated array of data labels, called the **index**.
- The **index** of a **Series** is an **array of labels** that correspond to the **values** in the **Series**. The **index** of a **Series** is an **optional parameter** that **defaults** to a sequence of integers starting at **zero**.
- The **Series** object is a **core data structure** in **Pandas**.

array numpy

index

0	5
1	8
2	9
3	7

Series

index

"Pepe"

"Pepe"

"Juan, la"

"Juan, to"

5		
8		
9		
7		



Querying a Series

- You could **query** a **Series** using **indexing** (boolean or fancy).
- You could **query** a **Series** using `loc` and `iloc` indexers.



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The “DataFrame” Data Structure

- A **DataFrame** is a **two-dimensional labeled data structure** with *columns* of potentially different types.
- A **DataFrame** is a **tabular data structure** that is *similar* to a spreadsheet or a SQL table.
- A **DataFrame** is a **core data structure** in **Pandas**. It is a two-dimensional size-mutable data structure with **labeled axes** (rows and columns).
- A **DataFrame** is a *container* for **Series** objects.



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DataFrame Indexing and Loading

- You could *index* a **DataFrame** using **column names**.
- You could *load* a **DataFrame** from a **CSV** file.
- You could *load* a **DataFrame** from a **JSON** file.
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DateTime Handling in Pandas

- You could *convert* a **string** to a **datetime** object using the `to_datetime()` method.
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Missing Values in a DataFrame

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- You could **fill** missing values in a **DataFrame**. The `fillna()` method returns a **DataFrame** with missing values filled.
- You could **drop** missing values in a **DataFrame**. The `dropna()` method returns a **DataFrame** with missing values dropped.



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- You could **drop** missing values in a **DataFrame**. The `dropna()` method returns a **DataFrame** with missing values dropped.



Merging DataFrames

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Pandas Idioms

- You could **apply functions** to **DataFrames**.
- You could **chain methods** in **Pandas**.



Pandas Idioms

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Outline

- 1 Foundations of Python
- 2 Numerical Analysis with Numpy
- 3 Data Manipulation with Pandas



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

